

RESEARCH ARTICLES

QUANTUM SIMULATION

Shapiro steps in strongly-interacting Fermi gases

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Driven many-body systems exhibit diverse and complex dynamical behaviors. Here, we report the observation of Shapiro steps in periodically driven Josephson junctions between strongly interacting Fermi superfluids of ultracold atoms. The height and the width of the observed quantized plateaus in the current-potential characteristics mirror the external drive frequency and the junction nonlinear response. Direct measurements of the current-phase relationship showcase how Shapiro steps arise from the synchronization between the relative phase of the two reservoirs and the external drive. Such a mechanism is further supported by the detection of periodic phase-slippage processes, in the form of vortex-antivortex pairs. Our results are corroborated by a circuital model and numerical simulations. Our work may open prospects for studying emergent nonequilibrium dynamics in quantum many-body systems under external drives.

Driven many-body platforms offer a powerful route to probe nonequilibrium quantum dynamics and emergent collective phenomena (1). A paradigmatic example is provided by Josephson junctions (JJs) under driving currents. A Josephson junction comprises two superconducting islands separated by a thin insulating barrier. Josephson showed that a dissipationless current I can flow across the junction, maintained solely by the phase difference ϕ between the two superconducting reservoirs (2). Above a critical current I_c , the junction transitions into the so-called voltage state: a finite voltage V develops across the junction, which is directly related to the phase difference dynamics as provided by the Josephson-Anderson relation $V = \frac{\hbar}{2e} \dot{\phi}$, with $\hbar$ being the reduced Planck constant and e the electron charge. In this regime, the Josephson current time evolution is characterized by the frequency $\omega_J = 2eV/\hbar$. The junction is thus an element that operates in cycles with a natural frequency controllably set by the externally injected current (3, 4).

When an AC drive with frequency ω is summed to a DC bias, the junction dynamics synchronize with the external drive, resulting in $\omega_J = n\omega$, with n a positive integer. This synchronization yields quantized voltage plateaus, known as Shapiro steps, appearing at $\langle V \rangle_t = n\hbar\omega/2e$ in the time-averaged voltage-current characteristics (5). Shapiro steps arise from a phase-locking process where ϕ advances by $2\pi n$ during each cycle of the driving field, undergoing n phase slip processes and

establishing coherence between the junction phase and the driven field. Since their observation in superconducting Josephson junctions (SJJs) (5), Shapiro steps have been investigated in various nonlinear systems, from charge density waves (6), colloidal systems (7, 8), superconducting nanowires (9) and nanotubes (10), and ^3He weak links (11). Shapiro steps in SJJs play a crucial role in metrology, in calibration within electronics, and in defining quantum standards (12, 13).

Recent years have witnessed considerable progress in developing and operating circuit architectures employing ultracold atoms (14). These atomtronics circuits feature enhanced control and flexibility of their working conditions that may go beyond conventional electronics. One of the most promising atomic platforms for implementing such circuits consists of ultracold Fermi gases (15). These systems provide a setting for different regimes of superfluidity, ranging from the Bardeen-Cooper-Schrieffer (BCS) limit of weakly bound fermion pairs to a Bose-Einstein condensate (BEC) of tightly bound molecules, including the intermediate universal, strongly correlated unitary regime (16, 17). In particular, unitary Fermi gases (UFGs) serve as a prototypical example of strongly interacting systems, sharing similarities with other strongly interacting Fermi systems found in nature, from nuclear matter to neutron stars (18). Research on quantum transport phenomena within these systems has explored a variety of configurations, including mesoscopic two-terminal setups (19), ring geometries (20, 21), optical lattices (22–25), and scenarios involving disorder (26) or periodic modulation of the harmonic trapping potential (27–31).

Josephson effects have been observed in both three-dimensional (32, 33) and two-dimensional (34) unitary superfluids, providing insights into the microscopic properties of these gases. However, despite various theoretical proposals (35–38), the observation of Shapiro steps in atomic Josephson junctions has remained elusive. Recently, Shapiro steps have been predicted to occur by periodically oscillating the position of a thin tunneling barrier within a weakly interacting BEC (39). However, for Fermi superfluids, it remains an open question whether strong interactions could affect the dynamics. Interactions are expected to enhance phase coherence across the junction and potentially reduce decoherence effects associated with the finite compressibility of atomic superfluids (32, 33), but their impact on synchronization processes is yet to be understood. Investigating these effects is crucial for deepening our understanding of strongly interacting systems and for revealing their microscopic behavior under external drives.

In this work, we report the observation of Shapiro steps in a periodically driven atomic Josephson junction connecting two strongly interacting unitary Fermi superfluids. Similar to SJJs, we find that the external driving frequency determines the height of Shapiro steps, whereas the driving amplitude affects their width. Moreover, we access a detailed microscopic understanding of the physical mechanism behind the Shapiro steps by operating the junction in the BEC regime and directly measuring its phase dynamics. By analyzing the current-phase relation under the external drive and observing the phase evolution across different Shapiro steps, we demonstrate the synchronization between the junction phase and the external drive. Both for a BEC and a unitary junction, we observe that the Shapiro dynamics are accompanied by vortex-antivortex pairs emissions, in agreement with the results of (39). Our results showcase the response of a many-body strongly interacting system to an external drive, resulting in synchronized dynamics.

Creating and driving the atomic junction

We produce strongly interacting Fermi superfluids of $N \sim 2 \times 10^4$ atom pairs by cooling a balanced mixture of the first and third lowest hyperfine states of ^6Li to temperatures $\sim 0.3(1) T_c$, where T_c is the superfluid critical temperature (40). Interparticle interactions are parameterized by $1/k_F a$, where $k_F = \sqrt{2mE_F}/\hbar$ is the Fermi wave

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vector, E_F the Fermi energy, and a the s -wave scattering length. We tune a in the vicinity of the Feshbach resonance at $690G$ (41), enabling the exploration of various superfluid regimes across the BEC-BCS crossover. In this work, we focus on two different interaction regimes: a UFG with Fermi energy $E_F/\hbar = 11.8(4)$ kHz, and a molecular BEC at $1/k_F a = 3.3(1)$ with $E_F/\hbar = 9.2(3)$ kHz. The gas is confined in the $x-y$ plane using a box-like potential with dimensions $L_x \times L_y$ of $125 \mu\text{m}$ by $17.5 \mu\text{m}$, and along the vertical z direction by a strong harmonic confinement (40). The trap configuration establishes a nearly homogeneous in-plane atomic density with the superfluid remaining in the three-dimensional regime. To engineer the atomic JJ, we superimpose a thin repulsive optical barrier onto the atomic sample, which can be displaced along the x axis (Fig. 1A). The barrier intensity profile and position are dynamically controlled by a digital micromirror device (DMD), whose designed pattern is projected onto the atomic plane using a high-resolution microscope objective (33, 42). The barrier features a full width at half maximum of $0.9(1) \mu\text{m}$ along the x direction (40), comparable to the healing length of the investigated superfluids, ranging between 0.6 and $1.7 \mu\text{m}$ for

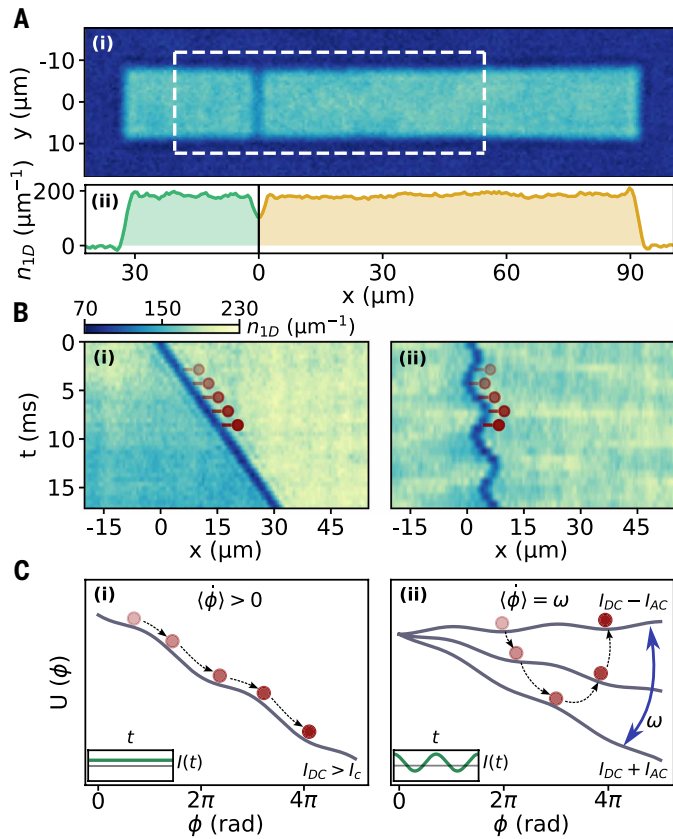


Fig. 1. Current injection in a homogeneous atomic JJ. (A) (i) In situ image of the JJ at unitarity, averaged over 15 repetitions. (ii) 1D density profile of the junction integrated along the y direction. (B) 1D density profiles as a function of time for (i) $v_{DC} = 1.5$ mm/s and $x_{AC} = 0$. (ii) $v_{DC} = 0.4$ mm/s and $x_{AC} = 2 \mu\text{m}$, modulating with a frequency of 175 Hz. Each row corresponds to the integrated average density profile over five experimental realizations. Red symbols of decreasing transparency mark the timing of the corresponding evolution illustrated in (C). (C) Sketch of the phase particle dynamics in the washboard potential for the conditions of (B). (i) The voltage state of the JJ corresponds to the particle rolling down the potential, causing a voltage increase. (ii) Introducing an alternating current modulates the tilt of the potential over time. For an overdamped JJ, the phase particle velocity locks to the external drive, $\langle \dot{\phi} \rangle_t = n\omega$, yielding to a synchronized motion that crosses n potential minima during each modulation cycle. The insets show the current for the two illustrated cases.

typical gas parameters. The barrier height, V_0 , is set to be larger than the gas chemical potential, μ_0 , with $V_0/\mu_0 = 1.3(1)$ for the unitary, and $1.2(1)$ for the BEC JJs, ensuring that the junction always operates in the tunneling regime (33, 42). Our working conditions are complementary to those reported in (43), where Shapiro steps are observed in a weakly interacting BEC junction in the weak link regime.

We inject currents into our junction by precisely moving the tunneling barrier along specific trajectories controlled with the DMD (33, 40, 42). The in-plane homogeneous density of the superfluid ensures that the injected current $I(t)$ is independent of its longitudinal position, and it is determined only by the instantaneous barrier velocity $v(t)$. The instantaneous current reads $I(t) = v(t)\tilde{n}L_y = v(t)N/L_x$ with $\tilde{n} = \int n_{3D}(z)dz = N/L_x L_y$ the planar density. To introduce a biased alternating current, we move the barrier according to the trajectory $x(t) = v_{DC}t + x_{AC}\sin(\omega t)$ (39), giving rise to the instantaneous injected current $I(t) = I_{DC} + I_{AC}\cos(\omega t)$, with $I_{DC} = v_{DC}N/L_x$, and $I_{AC} = \omega x_{AC}N/L_x$ (Fig. 1B). We explore the junction operation under different driving conditions by tuning the trajectory parameters v_{DC} , x_{AC} , and ω , as illustrated in Fig. 1B for bias (i) and composite current injections (ii). The junction dynamics can be portrayed with the well known washboard potential analogy (4), as sketched in Fig. 1C. The phase of the JJ follows the same equation of motion as that of a classical particle subject to a viscous force moving in a tilted washboard potential, where the tilt is given by the external current I . For a DC bias above I_c , the particle enters the running-phase regime, where $\langle \dot{\phi} \rangle_t \neq 0$, and the junction operates resistively [Fig. 1C (i)]. For an AC drive, in the overdamped regime, where friction overtakes inertia, the velocity locks to the driving tilt, resulting in a steady but synchronized descent, crossing n potential wells during each modulation cycle [Fig. 1C (ii)]. This resonant motion locks the mean particle velocity to fixed values $\langle \dot{\phi} \rangle_t = n\omega$, leading to distinct, equally spaced velocity steps, the Shapiro steps. In both cases, the descending movement of the phase particle produces a number of phase-slip events, synchronized with the drive in the AC case.

Shapiro steps in a unitary junction

We probe the response of our junction by measuring the $I - \Delta\mu$ characteristic curves as a function of the bias current, I_{DC} , after three driving periods T (40), where $T = 2\pi/\omega$. Here, $\Delta\mu = \mu_L - \mu_R$ represents the chemical potential difference between the left μ_L and right μ_R reservoirs measured from in situ density images. Curves for $I_{AC} = 0$ and $I_{AC} > 0$ for two example driving parameter sets are shown in Fig. 2, A and B. In the undriven scenario, we recover the hallmark behavior of a current-biased JJ, featuring a zero-imbalance plateau for DC currents below I_c (33). For the AC-driven JJ instead, we observe the emergence of distinct plateaus, with number, widths, and heights strongly depending on the driving parameters. To characterize the plateau properties, we perform a phenomenological fit of each step in the $I - \Delta\mu$ curve using multiple, patched sigmoidal functions (40), from which we get the height $\Delta\mu_n$ and the width ΔI_n of each step. The step height is in agreement with $\Delta\mu_n = n\hbar\omega$ for various driving frequencies and does not depend on I_{AC} (Fig. 2C): It is determined exclusively by the external driving frequency, whereas the driving strength I_{AC} affects only the position and width of each step (Fig. 2D).

To quantitatively describe the observed $I - \Delta\mu$ characteristics, we exploit the resistive and capacitively shunted junction (RCSJ) circual model (3, 4). In this model, the junction is set in parallel with resistive and capacitive components, with the total circulating current being

$$I(t) = I_c \sin\phi + \hbar G \dot{\phi} + \hbar C \ddot{\phi} \quad (1)$$

The supercurrent contribution is given by the sinusoidal current-phase relation $I_c \sin\phi$; the resistive and capacitive contributions are $G\dot{\phi}$ and $C\ddot{\phi}$, respectively. Similarly as for SJJ, the Josephson-Anderson relation $\Delta\mu = -\hbar\dot{\phi}$ connects the phase dynamics to the chemical potential build-up (35, 44), with $\Delta\mu$ playing the role of the junction voltage.

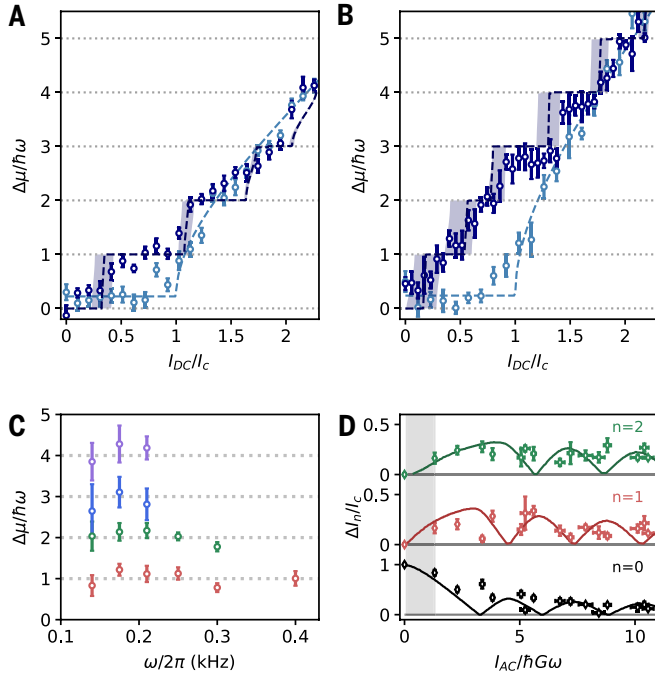


Fig. 2. Shapiro steps in strongly interacting Fermi superfluids at unitarity.

(A and B) Current-voltage characteristic for different AC drive settings (dark blue): $I_{AC}/I_c = 1.6(1)$ and $\omega = 2\pi \times 210$ Hz (A), and for $I_{AC}/I_c = 2.4(1)$ and $\omega = 2\pi \times 175$ Hz (B). In both panels, light blue symbols report the corresponding DC curve at $I_{AC}/I_c = 0$. Dashed light blue lines represent the fit with the stationary solution of the undriven overdamped RCSJ model (40); dark blue ones show the results for the driven scenario, where shades account for the fitting error on I_c . (C) Shapiro step height characterization. Colors label the 1st (red), 2nd (green), 3rd (blue), and 4th (purple) step. Dotted lines represent $\Delta\mu = n\hbar\omega$. Data points represent an average over at least three measurements at the same ω and different I_{AC} . (D) Step half-width characterization: Symbols report the measured step width for $\omega = 2\pi \times 175$ Hz, and variable I_{AC} . Error bars represent the error of the centroid of the multiple sigmoid fit. Solid lines correspond to the numerical solutions of the current-driven overdamped RCSJ model. G results from the fit of the DC curve (40). The gray shaded area marks the inaccessible parameter region.

The Stewart-McCumber parameter $\beta_c = I_c C / \hbar G^2$ determines the dynamic regime of the junction, discriminating between underdamped ($\beta_c \gg 1$) and overdamped ($\beta_c \ll 1$) (4). In the latter, the undriven solution is analytically given by $\Delta\mu = G^{-1} \sqrt{I_{DC}^2 - I_c^2}$, which we use to fit the DC measurements to extract I_c and G . By contrast, in the driven scenario, the $I - \Delta\mu$ characteristic is evaluated by numerically solving Eq. 1. As reported in Fig. 1, A and B, the overdamped solutions of the RCSJ model display the characteristic Shapiro steps matching the measured ones. Even though we estimate $\beta_c \approx 14(2)$ for the unitary superfluid, the overdamped RCSJ numerical solutions well capture the observed step height, owing to the robustness of the Shapiro effect. The precise determination of the nonlinear behavior of the step widths depends on the precise knowledge of the modulation protocol (40). However, the step half-widths obtained from the current-driven overdamped RCSJ model are in qualitative agreement with the experimentally measured values (Fig. 2D). Under voltage-driven modulation (3, 4, 11, 40), the half-widths of Shapiro steps can be found analytically, $\Delta I_n / I_c = |J_n(V_{ac}/\hbar\omega)|$, deviating quantitatively at small driving amplitudes with respect to current-driven modulation (45). Corrections to the RCSJ caused by compressibility effects can also influence its behavior (40).

Phase dynamics and synchronization

To link the Shapiro steps with the phase dynamics, we tune our atomic junction to operate in the BEC regime at $1/k_F a = 3.3(1)$. Here, the reduced

interactions allow us to leverage matter-wave interference between the expanding reservoirs to directly measure the relative phase ϕ (46), which cannot be accessed at unitarity because of the low contrast of interference fringes (40). For a BEC junction, we access both $\Delta\mu - I$ and $\phi - I$ characteristics (33), as reported in Fig. 3, A and B, for $\omega = 2\pi \times 175$ Hz, $I_{AC}/I_c = 0$, and $I_{AC}/I_c = 1.4(1)$. In the nonmodulated case for $I_{DC} < I_c$, we recover the expected DC Josephson trend (33), i.e., $\phi = \arcsin(I_{DC}/I_c)$. The measured ϕ shows larger error bars only for currents near and above I_c , attributed to the running phase in the voltage state (40). The driven scenario displays Shapiro steps with a diminished contrast as compared to the unitary junction, owing to the larger compressibility of the BEC superfluid (40). Nevertheless, we observe Shapiro steps in the $I - \Delta\mu$ curve throughout the same frequency range explored for the unitary junction (inset of Fig. 3A). The junction phase ϕ remains well defined within the plateau region even at finite $\Delta\mu$, presenting notable fluctuations only in the transition regions between plateaus (40). To support our findings, we compare the experimental results with numerical simulation performed with a dynamical classical-field method under similar experimental conditions (40, 47). The numerical simulations quantitatively agree with our results for $\Delta\mu - I$ and $\phi - I$ characteristics, capturing the smoothness of the Shapiro steps in the experimental data. The agreement with the numerical simulation suggests that the observed lower contrast of the plateaus in the BEC is inherent to our junction. In particular, both the experimental data and the numerical simulation results show a reduction in the Shapiro step height for large ω , when the modulation frequency approaches the vertical trap frequency, which is lower in the BEC regime.

We demonstrate the dynamical synchronization between the junction phase and the external driving by monitoring ϕ as a function of time in the 0th and 1st Shapiro steps (Fig. 3C); the 2nd Shapiro step is shown in fig. S3. The interference fringes show an oscillating behavior in both cases, highlighted by the measured relative phase ϕ (Fig. 3D), which showcases the phase-locking and synchronized dynamics induced by the current drive (Fig. 3E). Note that phase-locking effects have also been observed in a superfluid system in a Floquet-driven tilted double-well potential (48). Whereas in the 0th step, the phase oscillates in phase with the injected current without reaching the critical values for phase slips, in the 1st step ϕ reaches this critical limit once every cycle, except for the transient behavior of the first modulation, leading to periodic phase slips, visible as depletions in the integrated interferograms of Fig. 3C (ii) and marked by the red arrows. The phase-slippage process produces a clear spike in ϕ for both the dynamical classical-field simulation and overdamped RCSJ solution, associated with an increase in the error on the measured phase (40, 49).

Depending on the junction dimensionality, phase slips manifest as different topological excitations: solitons in 1D (4, 50), vortex-antivortex pairs in two dimensions (39), and vortex rings in three dimensions (49). The kind of excitations created is determined by which is energetically favorable (51), or arises from a multistep process such as soliton decay into different topological excitations (51, 52). In our BEC junction, we directly observe the presence of vortex-antivortex pairs as localized density depletions in the superfluid after a short time of flight (40) (Fig. 3F). The different junction geometry exploited in (43) leads instead to the emission of vortex rings during the Shapiro dynamics. For the 1st Shapiro step, we observe vortices generated on the near-left of the barrier (i-ii) that subsequently move toward the bulk of the reservoir (iii-iv). After an additional modulation period, a second vortex-antivortex pair is similarly emitted from the barrier region (v). The dynamical classical-field simulations corroborate this mechanism (figs. S7 and S8): At the end of the first cycle, the phase difference accumulated along the junction releases a solitonic excitation from the barrier, which then quickly decays into vortex-antivortex pairs, via snake instability (40, 51). The periodic emission of vortex-antivortex pairs in Shapiro steps is a proxy of the underlying synchronized phase

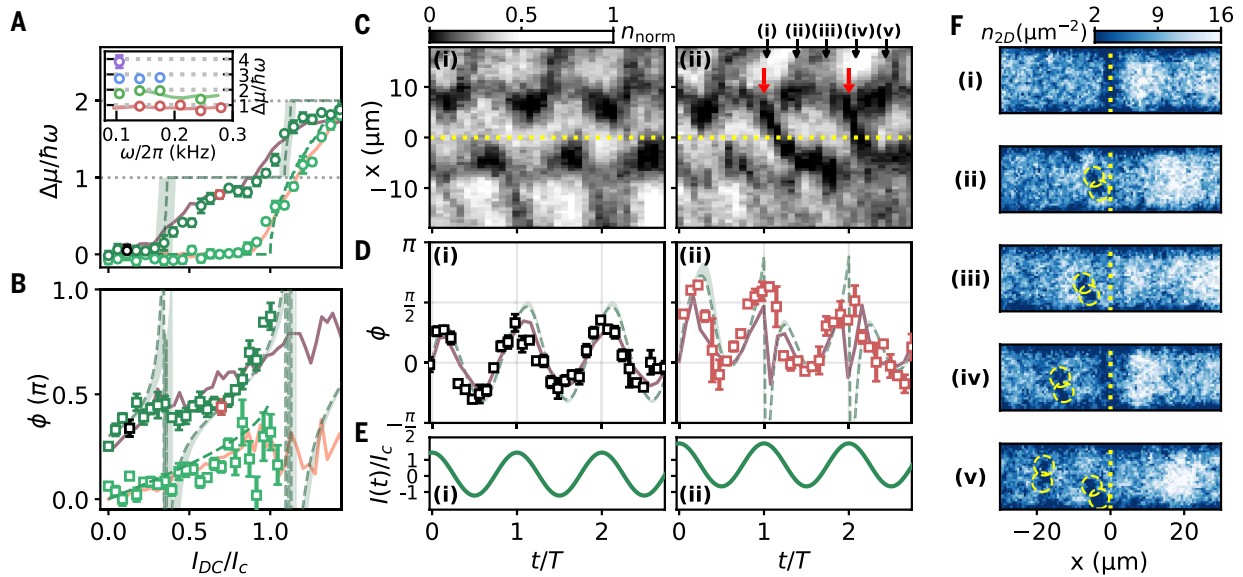


Fig. 3. Phase dynamics and dynamical locking in a driven atomic JJ in the BEC regime. (A) Current-voltage characteristic for a BEC junction under DC (light green symbols), and DC+AC drive at $\omega = 2\pi \times 175$ Hz and $I_{AC}/I_C = 1.4$ (1) (dark green symbols). Solid lines represent the numerical simulation results (see main text), with $I_{AC} = 0$ (orange) and $I_{AC}/I_C = 1.4$ (dark purple). Dashed lines represent the fit (light green) with the analytical and numerical solutions (dark green) of the overdamped RCSJ model (40). The shaded area accounts for the I_C fitting error. Inset: Shapiro step height characterization for a BEC junction. Symbols report the experimental step height obtained from the average over at least two repetitions plotted following the same color code as for Fig. 1C; dotted lines mark $\Delta\mu = \hbar\omega$; and the solid lines represent the step height obtained from numerical simulations. (B) Current-phase relation under same experimental conditions as in (A), vertically shifted by 0.3π for the DC+AC case. Solid and dashed lines represent the results of the numerical simulation and RCSJ model (40), respectively, as in (A). (C) Time evolution of the interference pattern between the reservoirs for $I_{AC}/I_C = 1.4$ (1) and $I_{DC}/I_C = 0.14$ (i), $I_{DC}/I_C = 0.83$ (ii), marked by the black and red symbols in (A) and (B). Each row reports the integrated fringe profile, averaged over five repetitions. Red arrows signal the occurrence of phase slips; the yellow dotted line marks the barrier position. (D) Measured phase-time evolution from a sinusoidal fit of the fringe patterns. Solid and dashed lines represent the results of the numerical simulation and RCSJ model, respectively. (E) Drive current under the same experimental conditions as in (C) and (D). (F) Single-shot density profiles in time-of-flight at $t/T = 1.03$ (i), 1.38 (ii), 1.73 (iii), 2.08 (iv), and 2.43 (v), showcasing the vortex emission process. Vortices are marked in yellow dashed circles. The yellow dotted line marks the barrier position.

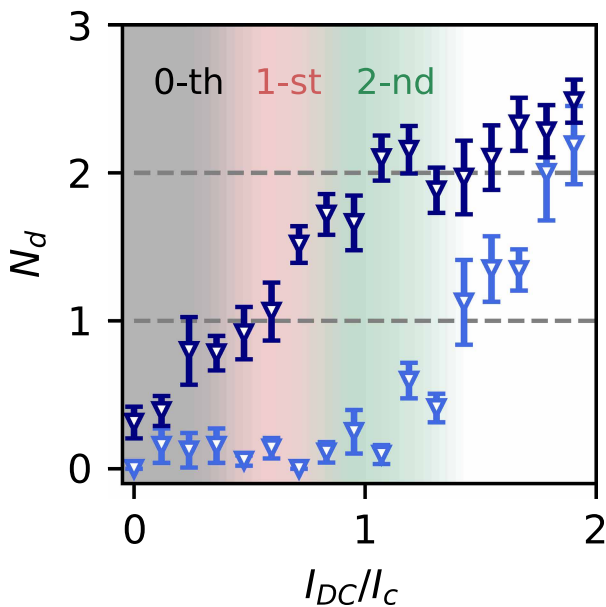


Fig. 4. Vortex-antivortex pairs as phase-slip proxies in a unitary JJ. Number of emitted vortex-antivortex pairs as a function of the bias current I_{DC}/I_C for $I_{AC}/I_C = 0$ (light blue), and $I_{AC}/I_C = 2.8$ (2) (dark blue). We measure the number of vortices after three periods of driving with a frequency $\omega = 2\pi \times 175$ Hz. The background color highlights the regions of the 0th, 1st, and 2nd Shapiro steps, with a color gradient reflecting the measured $\Delta\mu$ under the same driving conditions. Error bars represent the standard deviation of the mean over three to five realizations.

dynamics of the junction, which undergoes n phase-slippage processes in the n th step of each modulation cycle, as illustrated in the washboard potential analogy of Fig. 1C (ii) for $n = 1$.

As a direct measurement of the phase in the unitary regime is challenging (because of strong interactions limiting the fringe contrast), we probe the synchronization dynamics in the strongly interacting regime by detecting the number of emitted vortex pairs N_d as a function of the injected current (Fig. 4), revealing the occurrence of phase-slippage processes. In the nonmodulated case, vortices are observed to proliferate for $I_{DC} > I_C$, signaling a contribution to the junction conductance stemming from collective excitations, in agreement with previous observations in three-dimensional atomic JJs (42, 49). For the AC drive, the number of detected vortex pairs instead shows a step-like trend that closely resembles the Shapiro steps observed in the $I - \Delta\mu$ curve measured under the same driving conditions. In particular, in the 2nd Shapiro step, N_d doubles with respect to the 1st step, highlighting that, as a result of the underlying synchronized phase dynamics, the number of phase slips is proportional to the step number. The dissipative and resistive dynamics giving rise to the Shapiro steps in our atomic JJ are dominated by the collective excitations of the system, playing a role similar to that of quasiparticles produced by Cooper pair breaking in SJJ. Indeed, in all explored configurations, the junction always works in the regime of $\Delta\mu \ll \Delta$, with Δ being the superfluid gap, where broken pairs are energetically suppressed and the only accessible excitations are collective ones—namely, phonons, solitons and vortices (42).

Discussion and outlook

We have observed Shapiro steps in a current-driven atomic JJ composed of two weakly coupled strongly interacting Fermi gases. Shapiro steps appear as quantized plateaus in the relative chemical potential—the

analog of the voltage drop in SJJ—occurring at integer multiples of the external driving frequency. As the driving amplitude increases, the widths of these plateaus exhibit a nonlinear dependence, arising from the interplay between the external drive and the nonlinear features of the atomic JJ. We probe the synchronization process at the heart of Shapiro steps by directly accessing the microscopic phase dynamics of the junction, featuring periodic phase-slippage processes that manifest as vortex-antivortex pairs. We note that such a direct microscopic observation is challenging to carry out with the available techniques in solid-state driven JJs, which have several-orders-of-magnitude faster dynamics.

At a fundamental level, our results reflect the interplay between quantum phase coherence and dissipation dynamics in a strongly correlated Fermi system. Our work opens prospects for a deeper understanding of how quasiparticles and quantum phase slips contribute to quantum transport, such as that within driven superconducting networks (53–55).

Our driven atomtronic circuit can be exploited to investigate driven atomic JJ arrays, whether arranged linearly or in annular configurations (56). In turn, this opens the way to simulate important paradigms in quantum synchronization, as provided by Kuramoto-like models (57), by driven atomic JJs in the presence of strong interactions and with disordered couplings between the links (58, 59).

In addition, as Shapiro steps provide the voltage standards in quantum electronics (13), we could leverage the quantized steps to access a direct measurement of chemical potential differences (36). This approach would be specifically valuable in the intermediate regimes of the cross-over superfluids, where the equation of state is challenging (16).

REFERENCES AND NOTES

1. A. Pikovsky, M. Rosenblum, J. Kurths, *Synchronization: A Universal Concept in Nonlinear Sciences* (Cambridge Univ. Press, 2001).
2. B. Josephson, *Phys. Lett.* **1**, 251–253 (1962).
3. A. Barone, G. Paternò, *Physics and Applications of the Josephson Effect* (Wiley, 1982).
4. M. Tinkham, *Introduction to Superconductivity*, Dover Books on Physics (Dover Publications, ed. 2, 2004).
5. S. Shapiro, *Phys. Rev. Lett.* **11**, 80–82 (1963).
6. G. Grüner, *Rev. Mod. Phys.* **60**, 1129–1181 (1988).
7. M. P. Juniper, A. V. Straube, R. Besseling, D. G. Aarts, R. P. Dullens, *Nat. Commun.* **6**, 7187 (2015).
8. A. P. Stikuts, S. Mishra, A. Ryabov, P. Maass, P. Tierno, *Nat. Commun.* **16**, 2966 (2025).
9. R. C. Dinsmore III, M.-H. Bae, A. Bezryadin, *Appl. Phys. Lett.* **93**, 192505 (2008).
10. I. Bogush, V. M. Fomin, O. V. Dobrovolskiy, *Phys. Rev. B* **111**, 214510 (2025).
11. R. W. Simmonds, A. Marchenkov, J. C. Davis, R. E. Packard, *Phys. Rev. Lett.* **87**, 035301 (2001).
12. B. Jeanneret, S. P. Benz, *Eur. Phys. J. Spec. Top.* **172**, 181–206 (2009).
13. J. Kohlmann, R. Behr, *Development of Josephson Voltage Standards* (InTech, 2011).
14. L. Amico et al., *Rev. Mod. Phys.* **94**, 041001 (2022).
15. J. Polo et al., *Quantum Sci. Technol.* **9**, 030501 (2024).
16. W. Ketterle, M. W. Zwerlein, *Riv. Nuovo Cim.* **31**, 247–422 (2008).
17. W. Zwerger, Ed., *The BCS-BEC Crossover and the Unitary Fermi Gas* (Springer, 2012).
18. M. W. Zwerlein, *Novel Superfluids*, vol. 2 (Oxford University Press, Oxford, 2014).
19. S. Krinner, T. Esslinger, J.-P. Brantut, *J. Phys. Condens. Matter* **29**, 343003 (2017).
20. Y. Cai, D. G. Allman, P. Sabharwal, K. C. Wright, *Phys. Rev. Lett.* **128**, 150401 (2022).
21. G. Del Pace et al., *Phys. Rev. X* **12**, 041037 (2022).
22. P. T. Brown et al., *Science* **363**, 379–382 (2019).
23. W. Xu, W. R. McGehee, W. N. Morong, B. DeMarco, *Nat. Commun.* **10**, 1588 (2019).
24. M. A. Nichols et al., *Science* **363**, 383–387 (2019).
25. M. Lebrat et al., *Phys. Rev. X* **8**, 011053 (2018).
26. M. Schreiber et al., *Science* **349**, 842–845 (2015).
27. R. Anderson et al., *Phys. Rev. Lett.* **122**, 153602 (2019).
28. P. B. Patel et al., *Science* **370**, 1222–1226 (2020).

29. D. Hernández-Rajkov et al., *New J. Phys.* **23**, 103038 (2021).
30. Z. Yan et al., *Science* **383**, 629–633 (2024).
31. C. Cabrera et al., *Phys. Rev. Res.* **7**, L032018 (2025).
32. G. Valtolina et al., *Science* **350**, 1505–1508 (2015).
33. W. J. Kwon et al., *Science* **369**, 84–88 (2020).
34. N. Luick et al., *Science* **369**, 89–91 (2020).
35. A. Smerzi, S. Fantoni, S. Giovanazzi, S. R. Shenoy, *Phys. Rev. Lett.* **79**, 4950–4953 (1997).
36. S. Kohler, F. Sols, *New J. Phys.* **5**, 94–94 (2003).
37. A. Eckardt, T. Jinasundera, C. Weiss, M. Holthaus, *Phys. Rev. Lett.* **95**, 200401 (2005).
38. J. Grond et al., *New J. Phys.* **13**, 065026 (2011).
39. V. P. Singh, J. Polo, L. Mathey, L. Amico, *Phys. Rev. Lett.* **133**, 093401 (2024).
40. See supplementary materials.
41. G. Zürn et al., *Phys. Rev. Lett.* **110**, 135301 (2013).
42. G. Del Pace, W. J. Kwon, M. Zaccanti, G. Roati, F. Scazza, *Phys. Rev. Lett.* **126**, 055301 (2021).
43. E. Bernhart et al., *Science* **390**, 1130–1133 (2025).
44. F. Meier, W. Zwerger, *Phys. Rev. A* **64**, 033610 (2001).
45. R. Panghotra et al., *Commun. Phys.* **3**, 53 (2020).
46. C. Li, Q. Liang, P. Paranjape, R. Wu, J. Schmiedmayer, *Phys. Rev. Res.* **6**, 023217 (2024).
47. V. P. Singh et al., *Phys. Rev. A* **93**, 023634 (2016).
48. S.-C. Ji et al., *Phys. Rev. Lett.* **129**, 080402 (2022).
49. A. Burchianti et al., *Phys. Rev. Lett.* **120**, 025302 (2018).
50. F. Binanti, K. Furutani, L. Salasnich, *Phys. Rev. A* **103**, 063309 (2021).
51. W. Van Alphen, H. Takeuchi, J. Tempere, *Phys. Rev. A* **100**, 023628 (2019).
52. M. J. Ku, B. Mukherjee, T. Yefsah, M. W. Zwerlein, *Phys. Rev. Lett.* **116**, 045304 (2016).
53. W. Guichard, F. W. Hekking, *Phys. Rev. B Condens. Matter Mater. Phys.* **81**, 064508 (2010).
54. K. Y. Arutyunov, D. S. Golubev, A. D. Zaikin, *Phys. Rep.* **464**, 1–70 (2008).
55. I. M. Pop et al., *Nat. Phys.* **6**, 589–592 (2010).
56. L. Pezzè et al., *Nat. Commun.* **15**, 4831 (2024).
57. Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence* (Springer Berlin Heidelberg, 1984).
58. D. J. Watts, S. H. Strogatz, *Nature* **393**, 440–442 (1998).
59. B. R. Trees, V. Saranathan, D. Stroud, *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* **71**, 016215 (2005).
60. G. Del Pace et al., Shapiro steps in strongly-interacting Fermi gases [Data set], Zenodo (2025); <https://doi.org/10.5281/zenodo.17303804>.

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Competing interests: The authors declare that they have no competing interests. **Data and materials availability:** The study involved no new materials preparation. All experimental data and numerical simulation codes shown in this work are available on Zenodo (60).

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SUPPLEMENTARY MATERIALS

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Shapiro steps in strongly-interacting Fermi gases

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Editor's summary

In the Josephson effect, a current can flow through an insulating barrier between two superconductors without dissipation. For a high enough current, dissipation does occur, and if microwave radiation is also applied, a series of plateaus—the so-called Shapiro steps—appear in the current-voltage characteristic. Instead of using superconductors, Del Pace *et al.* and Bernhart *et al.* observed Shapiro steps in an atomic Josephson junction. The researchers moved a barrier through an ultracold gas at a speed that had a periodically modulated component to emulate microwave radiation. The height of the resulting Shapiro steps in the chemical potential difference was proportional to the frequency of the modulation. —Jelena Stajic

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